Hyperspectral Image Compression Using Sampling and Implicit Neural Representations

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***Abstract*— Hyperspectral images record the electromagnetic spectrum, and each hyperspectral pixel often stores hundreds of channels. Consequently, a hyperspectral image contains an order of magnitude more information than a similar-sized RGB color image. Concomitant with the decreasing cost of capturing these images, there is a need to develop efficient techniques for storing, transmitting, and analyzing hyperspectral images. This article develops a method for hyperspectral image compression using implicit neural representations (INRs) where a multilayer perceptron (MLP) network with sinusoidal activation functions "learns" to map pixel locations to pixel spectrum for a given hyperspectral image. This representation, thus, acts as a compressed encoding of this image, and the original image is reconstructed by evaluating this network at each pixel location. We introduce a sampling scheme to achieve better compression times while keeping decoding errors low. The proposed method is evaluated on four benchmarks against 16 other schemes for hyperspectral compression, and according to the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) metrics, the method developed in this article achieves state-of-the-art compression rates at low-bit rates. In addition, we show that the proposed sampling technique reduces encoding times.**

***Index Terms*— Hyperspectral image compression, implicit neural representations (INRs).**

I. Introduction

UNLIKE a gray-scale image that records a single intensity value or a color image that often records three values at each pixel, hyperspectral images capture electromagnetic spectrum [1], [2]. Therefore, each pixel in a hyperspectral image contains tens or hundreds of values, representing recorded reflectance at various frequency bands. As a result, hyperspectral images offer greater possibilities for object detection, material identification, and scene analysis than those provided by gray-scale or color images. The costs associated with capturing high-resolution (both spatial and spectral) hyperspectral images continue to decrease, and it is no surprise that hyperspectral images have found widespread use in areas such as remote sensing, biotechnology, crop analysis, environmental monitoring, food production, medical diagnosis, pharmaceutical industry, mining, and oil and gas exploration [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16]. Hyperspectral images require many orders of magnitude more space than what is needed to store a similarly sized gray-scale or color image. Consequently, there is a need to develop efficient schemes for capturing, storing, transmitting, and analyzing hyperspectral images. This work studies the problem of hyperspectral image compression, noting that hyperspectral image compression plays an important role in the storage and transmission of these images.

Recently, there has been a surge in interest in learning-based compression schemes. For example, autoencoders [17] and rate-distortion autoencoders [18], [19] have been used to learn compact representations of the input signals. Here, network weights together with the signal signature—latent representation in the case of autoencoders—serve as the compressed representation of the input signal. Other concurrent works are exploring the use of implicit neural representations (INRs) for signal compression. INRs are particularly well-suited to describe data that live on an underlying grid. These are able to capture complex patterns and connections between the data without the need for explicit parameterization of the grid structure. Specifically, INRs learn a mapping between the grid coordinates and the related data values (e.g., a mapping between pixel location to its spectrum) [20]. Mathematically, consider a hyperspectral image $\mathbf{I} \in \mathbb{R}^{h \times w \times c}$, where h, w, and c denote the height, width, and the number of channels, respectively. The goal is to learn a function $f_\Theta : (x, y) \mapsto \mathbf{I}[x, y]$, where (x, y) denote pixel coordinates and $\mathbf{I}[x, y] \in \mathbb{R}^c$ denotes the pixel spectrum. Θ, which serves as an encoding for the image $\mathbf{I}$, denotes the function parameters. The original image can be reconstructed given Θ by evaluating f_Θ at $[1, h] \times [1, w]$.

Using INRs to "encode" hyperspectral images offers the following advantages.

1) *Flexibility:* INRs can accurately represent complex and irregular grids or surfaces. Contrary to explicit grid-based representations, which require the explicit definition of the grid structure, implicit representations can adjust to the data without imposing strict grid limitations.
2) *Generalization Ability:* INRs exhibit strong generalization capabilities to previously unknown data items. They have the ability to catch complex patterns in the data, enabling extrapolation beyond the recorded grid points. This is especially advantageous when working with data that are sparsely or irregularly collected.
3) *Computational Efficiency and Scalability:* Implicit representations can offer computational efficiency, particularly when dealing with high-dimensional data

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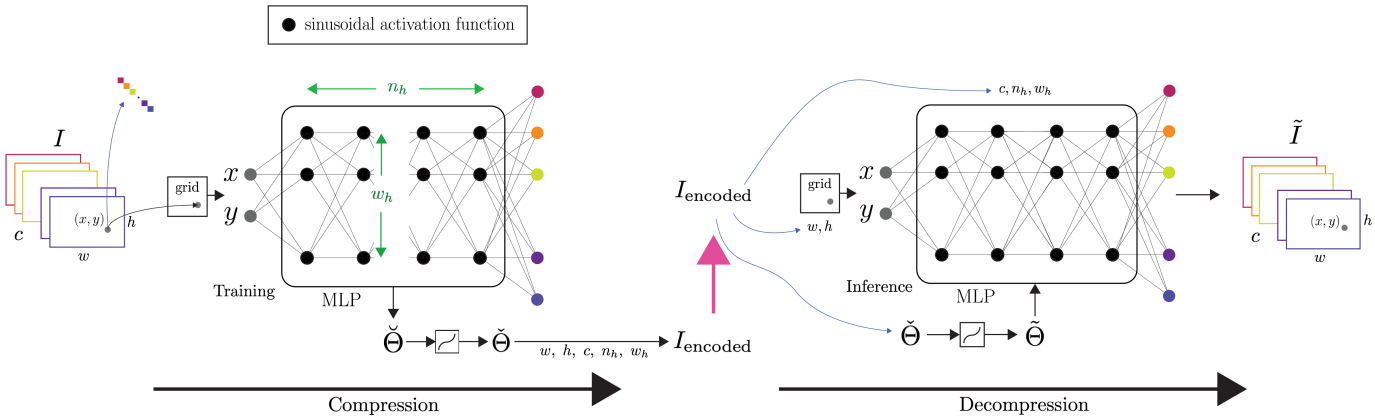


Fig. 1. Compression and decompression pipeline. (Left) An MLP with a periodic activation function is *trained* to map pixel locations to the pixel's spectral signature. (Right) Once fit, MLP is used to reconstruct the hyperspectral image by performing *inference* at various pixel locations.

or large grid sizes. Instead of explicitly storing or computing values for each grid point, they can generate values on-demand by evaluating neural networks. In addition, implicit representations provide excellent scalability in high-dimensional spaces. The complexity of explicit grid-based approaches frequently increases dramatically as the dimensionality of the underlying grid increases. Implicit representations, on the other hand, are capable of properly managing data with a large number of dimensions.

4) *Smoothness and Continuity Assumptions:* INRs have the ability to depict smooth and uninterrupted changes in the data accurately. Describing phenomena that demonstrate progressive variations throughout the grid is advantageous because neural networks naturally smooth out the representations they learn.
5) *Learnable Representations:* Implicit representations can be learned directly from data. Neural networks can be trained using optimization approaches to accurately capture the underlying patterns in the grid-based data, enabling the identification of intricate relationships and features. In a nutshell, INRs provide a robust framework for encoding data that lie on a grid, offering flexibility, the ability to be generalized, computing efficiency, scalability, and the capacity to capture smooth and continuous variations in the data.

We investigate the use of INRs for hyperspectral image compression and show that it is possible to achieve high rates of compression while maintaining acceptable peak signal-to-noise ratio (PSNR) values. Fig. 1 provides an overview of the proposed compression and decompression pipeline. We evaluate the proposed approach on four benchmarks (see Fig. 2)—1) Indian Pines; 2) Jasper Ridge; 3) Pavia University; and 4) Cuprite—and show that at comparable bits-per-pixel-per-band (bpppb) values, our method achieves better PSNR values than those posted by existing hyperspectral compression methods. Specifically, we evaluate the proposed approach against: 1) JPEG [21], [22]; 2) JPEG2000 [23]; 3) PCA-DCT [24]; 4) PCA-JPEG2000 [25], [26], [27]; 5) MPEG [28]; 6) X264 [25], [26], [27]; 7) X265 [25], [26], [27]; 8) PCA-X264 [25], [26], [27]; 9) PCA-X265 [25], [26], [27]; 10) FPCA-JPEG2000 [29]; 11) 3D-DCT [30]; 12) 3D-DWT-SVR [31]; 13) WSRC [32]; 14) HEVC [33]; 15) RPM [34]; 16) 3D-SPECK [35]; and 17) an autoencoder-based scheme proposed in [36]. This list includes both the so-called classical approaches and the more recent learning-based methods. In addition, we propose a sampling technique to speed up the encoding process and show that sampling achieves faster compression times while achieving PSNR values similar to those obtained when the image is encoded without sampling.[1]

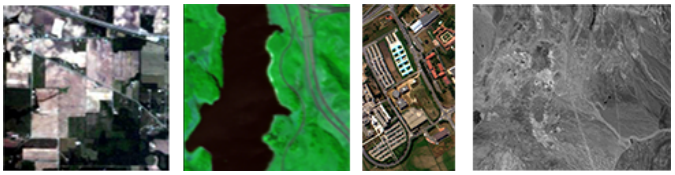


Fig. 2. Datasets used in this study (shown in pseudocolor). (L2R) Indian Pines, Jasper Ridge, Pavia University, and Cuprite.

The rest of this article is organized as follows. We discuss the related work in Section II. Section III describes the proposed method along with evaluation metrics. Datasets, experimental setup, and compression results are discussed in Sections IV and V. Section VI concludes this article with a summary and possible directions for future work.

II. Related Work

Hyperspectral images exhibit both spatial and spectral redundancies that can be exploited to achieve compression. Lossless compression schemes, e.g., those that use quantization or rely upon entropy coding, where it is possible to recover the original signal precisely, often do not yield large savings in terms of memory required to store or transmit a hyperspectral image [37], [38]. Lossy compression schemes,

[1]This article includes five variations of the proposed scheme. We use the following naming convention to refer to these methods: "ours" obviously refers to the methods developed in this work; "sampling" denotes whether image coordinates were sampled during the training procedure; and "8, 16, or 32 bits" refer to the number of bits used to store individual weights of the neural network.

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on the other hand, promise large savings while maintaining acceptable reconstruction quality. Interband compression techniques aim to eliminate spectral redundancy [39], while intraband compression techniques aim to exploit spatial correlations. Intraband compression techniques often follow the ideas developed for color image compression. Zhao et al. [40] exploit the fact that groups of pixels that are around the same location in two adjacent bands are strongly correlated and propose a scheme that performs both interband and intraband compressions. Principal component analysis (PCA) is a popular dimensionality reduction technique that has been used for hyperspectral image compression. PCA offers strong spectral decorrelation, and it is often employed to reduce the number of channels in a hyperspectral image. The remaining channels are subsequently compressed using Joint Picture Expert Group (JPEG) or JPEG2000 standard [23], [41], [42], [43].

Tensor decomposition methods have also been applied to the problem of hyperspectral image compression [44]. Tensor decomposition achieves dimensionality reduction while maintaining the spatial structure. Transform coding schemes that achieve image compression by reducing spatial correlation have also been used to compress hyperspectral data. Discrete cosine transform (DCT) has been used to perform intraband compression; however, it ignores interband (or spectral) redundancy. 3D-DCT that divides a hyperspectral image into $8 \times 8 \times 8$ datacubes is proposed to achieve both interband and intraband compressions [22], [30]. Similar to JPEG, which uses 8×8 blocks, 3D-DCT exhibits blocking effects in reconstructed hyperspectral images. The blocking effects can be removed to some extent by using wavelet transform instead [45], [46].

Video coding-based methods that treat each channel of a hyperspectral image as a frame in a video have also been used to perform hyperspectral image compression. Examples of these methods are MPEG, X264, X265, PCA-X264, and PCA-X265. These models rely upon interband spectral prediction to compress a hyperspectral image. This is similar to how interframe motion prediction is used for video encoding. The High-Efficiency Video Coding scheme uses (surface) Reflectance Prediction Modeling to achieve a high level of compression [34].

As mentioned earlier, recently, there has been a lot of interest in developing learning-based approaches for signal representation and compression. For example, autoencoder-based techniques have been proposed to compress hyperspectral images [18], [19], [36]. Hierarchical variational autoencoders have also been used for the purposes of hyperspectral image compression. Here, the latent variables are discretized for entropy encoding purposes [47], [48], [49] to achieve further savings. Our work also employs a similar strategy and uses quantization to reduce the space needed to store the model parameters θ. WSRC [32] uses a downsampling scheme that preserves relevant information. Super-resolution is used to maintain a high-frequency signal when reconstructing the original image from its compressed, low-resolution version. The three-dimensional DWT-SVR [31] scheme represents an image using 3-D wavelet coefficients that are compressed via regression using a support vector machine.

Work in the area of INRs has shown that it is possible to represent a signal by overfitting an appropriately designed neural network to it. Here, the parameters (weights) of the neural network serve as the compact representation of the signal, and it is possible to reconstruct the original signal by sampling the neural network at various input locations [50], [51], [52], [53], [54], [55], [56]. [57] shows that INRs with periodic activation functions are able to represent signals, including images, with high fidelity. This work serves as an inspiration for our work.

Similar to previous research on latent variable models [58], [59], [60], numerous studies [61], [62], [63] make an effort to close the amortization gap [64] by combining the usage of amortized inference networks with iterative gradient-based optimization procedures. Using inference time per instance optimization, [63] also identifies and makes an effort to bridge the discretization gap caused by quantizing the latent variables. The concept of per-instance model optimization is expanded upon in [65], which fine-tunes the decoder for each instance and transmits updates to the quantized decoder's parameters together with the latent code to provide better rate-distortion performance.

III. Image Compression Using INRs

Let us consider a w-by-h gray-scale image. We can represent this image as a function

$$I_{\text{grayscale}} : U \mapsto [0, 1]$$

where

$$U = \{(1, 1), \ldots, (w, 1), \ldots, (1, h), \ldots, (w, h)\}.$$

This notation captures the intuition that an image is a function over a 2-D grid that defines the pixel locations. The intensity at each pixel is then $I_{\text{grayscale}}(x, y)$. It is straightforward to extend this notation to hyperspectral image I as follows:

$$I = \begin{pmatrix} I_1 : U \mapsto [0, 1] \\ \vdots \\ I_c : U \mapsto [0, 1] \end{pmatrix}. \tag{1}$$

Here, for the sake of simplicity, we assume a w-by-h hyperspectral image comprising c channels. Using this notation, we can find the spectral signature of the pixel at location $(x, y) \in U$ as follows:

$$\{I_1(x, y), \ldots, I_c(x, y)\}.$$

Given this setup, it is possible to imagine a neural network that models the functions $I_1, \ldots, I_c$. Specifically, work on INRs suggests using a multilayer perceptron (MLP) with periodic activation functions to represent functions of the form shown in (1). Consider an MLP f_Θ with parameters Θ that maps locations in U to pixel spectral signatures

$$f_\Theta : U \mapsto \{I_1, \ldots, I_c\}.$$

Under this regime, training can be defined as

$$\breve{\Theta} = \arg\min_{\Theta} \mathcal{L}(I, f_\Theta)$$

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TABLE I
DISK LAYOUT FOR I_{encoded}. HERE, q DENOTES IF PARAMETERS Θ WERE QUANTIZED AT COMPRESSION TIME. bpp (OR #BITS-PER-PARAMETER) IS EITHER 32 OR 16

	w	h	c	n_h	w_h	q	Θ
#bits	16	16	16	8	8	1	bpp $\times$ n_Θ

where $\mathcal{L}$ is a loss function that is differentiable and captures the error between the original hyperspectral image and the decompressed hyperspectral image. We use the mean square error (MSE) to compute this loss. Given $\breve{\Theta}$, it is possible to reconstruct the original image I by sampling $f_{\breve{\Theta}}$ at the relevant locations. Parameters $\breve{\Theta}$, along with w, h, and c, plus the structure of the MLP, represent an encoding I_{encoded} of the hyperspectral image I that was used to train the MLP f_Θ. It is expected that the memory required to store I_{encoded} is an order of magnitude less than the memory needed to store the hyperspectral image. Table I shows information contained in I_{encoded}. The decompression process requires constructing the sampling locations U, setting up MLP f_Θ and initializing its weights to $\breve{\Theta}$, and evaluating $f_{\breve{\Theta}}$ at locations in U.

A. Metrics for Measuring Compression Quality

Similar to previous studies, we use PSNR to compare the proposed method with current schemes for hyperspectral image compression. PSNR, measured in *decibels*, is a frequently used metric in image compression. It measures the difference in "quality" between the original image and its compressed version. Higher PSNR values suggest that the compressed image is more similar to the original image, i.e., the compressed image preserves more of the information present in the original image, and it has higher quality. In addition, we also compare the compressed image using MSE, which computes the cumulative squared error between the original image and its compressed version. Lower values of MSE mean better reconstruction quality.

MSE is computed as follows:

$$\text{MSE} = \left\| I[i] - \tilde{I}[i] \right\| \tag{2}$$

where $\tilde{I}$ denotes the reconstructed image. MSE is used to calculate PSNR

$$\text{PSNR} = 10 \log_{10} \left(\frac{R^2}{MSE} \right) \tag{3}$$

where R is the largest variation in the input image in the previous equation. For instance, R is 1 if the input image is of the double-precision floating-point data. R is 255, for instance, if pixel values are stored as 8-bit unsigned integers.

Another metric that is commonly used for image compression studies is the structural similarity index (SSIM) [66]. SSIM accounts for the structural information that the human visual system is naturally attuned to. Therefore, SSIM is perceptually more meaningful then MSE, which treats pixels independently of each other. SSIM metric combines luminance, contrast, and structure, and the channelwise SSIM is computed as follows:

$$\text{SSIM}_{\text{channelwise}} = \frac{(2\mu_I \mu_{\tilde{I}} + C_1)(2\sigma_{I\tilde{I}} + C_2)}{\left(\mu_I^2 + \mu_{\tilde{I}}^2 + C_1\right)\left(\sigma_I^2 + \sigma_{\tilde{I}}^2 + C_2\right)} \tag{4}$$

where μ_I and $\mu_{\tilde{I}}$ are mean intensity values for the original image and its reconstruction, respectively. Similarly, σ_I^2 is the variance of the original image, and $\sigma_{I\tilde{I}}$ is the covariance value for the original image and its reconstruction. $C_1 = (k_1 \cdot L)^2$ and $C_2 = (k_2 \cdot L)^2$ are there to address division by a weak denominator. L, here, denotes the dynamic range of a pixel, and $K_1 = 0.01$ and $K_2 = 0.03$. The dynamic range L is typically expressed as $2^{(\text{bits per pixel})}$. The mean SSIM value is computed by averaging channelwise SSIM values. SSIM values range between 0 and 1, and larger SSIM values indicate a better reconstruction.

Bits-per-pixel-per-band (bpppb) is sometimes used to capture the level of compression. Recall bpppb for an uncompressed hyperspectral image is typically either 8 or 32, depending on how the pixel values are stored. Recall that hyperspectral pixels are often stored as 32-bit floats. For our model, bpppb is calculated as follows:

$$\text{bpppb} = \frac{\#\text{parameters} \times (\text{bits per parameter})}{(\text{pixels per band}) \times \#\text{bands}}. \tag{5}$$

Fig. 3 plots PSNR versus bpppb for the four datasets that we are using in this work. These plots confirm our intuition that higher bpppb leads to better compression quality as measured by PSNR values. bpppb calculations do not include the storage required to keep network structure (number of layers and the layer width) and the image information (height, width, and the number of channels or bands) needed to decode the image.

B. Compression Pipeline

The proposed compression method consists of two steps. Step 1 performs an architecture search (see Fig. 4). The goal here is to find an MLP that achieves the highest reconstruction accuracy for a given bpppb budget. Architecture search is performed by overfitting multiple MLPs having different numbers of hidden layers and hidden layers' widths to the hyperspectral image. Architecture search, however, means longer compression times. Step 2 involves quantizing and storing the parameters of the overfitted MLP to disk. Quantizing network parameters is desirable and results in further savings; however, this may further reduce the quality of the reconstructed image. This article includes results when network parameters are quantized using 16 and 8 bits.

1) Overfitting a SIREN Network: The compression procedure comprises overfitting a SIREN network f_Θ to a hyperspectral image I [57]. The width w and height h of the hyperspectral image are used to set up an input location grid on $[-1, +1] \times [-1, +1]$, and the MLP is trained to reconstruct a pixel's spectral signature given its location. The parameters $\breve{\Theta}$ of this overfitted MLP are quantized $\breve{\Theta}$. MLP structure that contains the number of hidden layers n_h, widths of these layers w_h, and the width w, height h, and the number of channels c of the original hyperspectral image I along with $\breve{\Theta}$ serves as a compressed encoding I_{encoded} of the hyperspectral image I (see Table I). Parameters $\breve{\Theta}$ are typically stored as 32-bit

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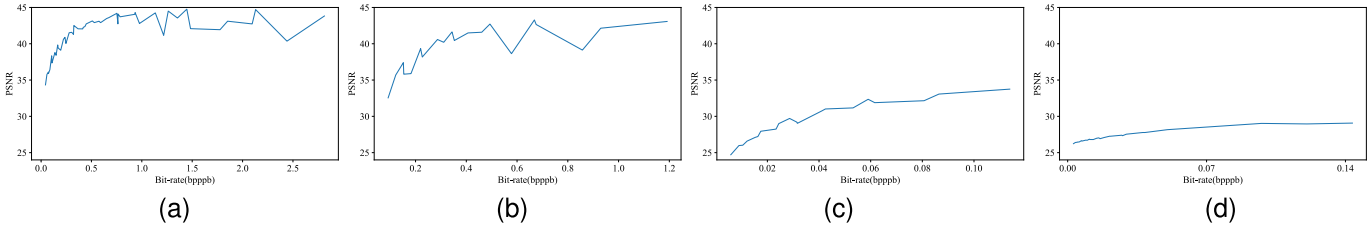


Fig. 3. Model capacity. PSNR versus bpppb. The trend of these plots confirms our intuition that PSNR values increase as bpppb numbers are increased. The plots are not monotonically nondecreasing. This has to do with the stochastic nature of MLP overfitting. (a) Indian Pines. (b) Jasper Ridge. (c) Pavia University. (d) Cuprite.

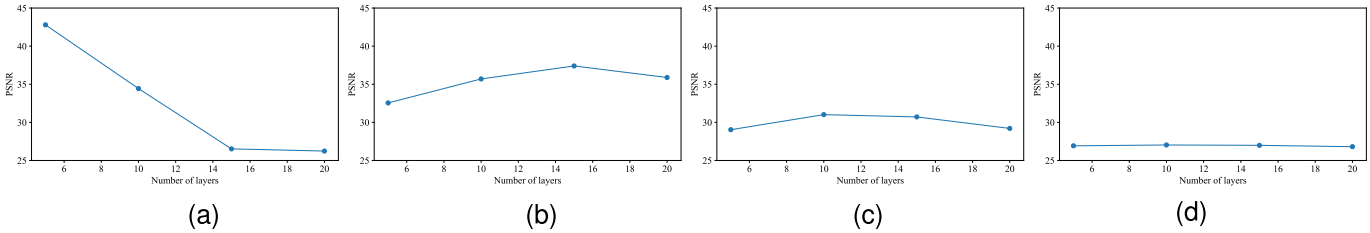


Fig. 4. Architecture search. Exploring MLP structure that achieves the best PSNR for different datasets (for a fixed bpppb budget). For our purposes, the MLP structure is defined by the number of hidden layers and the width of these layers. Together, the number and width of the hidden layers define network capacity. Note that bpppb is fixed for each dataset. This means that as the number of layers is increased, the width of these layers must decrease accordingly. This figure confirms that architecture search plays an important role. For example, the Indian Pines dataset is best compressed (as measured using PSNR) when using two layers, whereas the best MLP model for the Japster Ridge dataset contains 15 layers. (a) Indian Pines. (b) Jasper Ridge. (c) Pavia University. (d) Cuprite.

floats. However, further savings are possible by quantizing these parameters and storing them using fewer bits. We include compression results for when these parameters are stored as 16- and 8-bit values. The overfitted MLP contains

$$(w_h \times 2) + (w_h \times w_h)^{(n_h-1)} + (w_h \times c) \tag{6}$$

parameters.

C. Decompressing $I_{encoded}$

The hyperspectral image is reconstructed from its compressed encoding I_{encoded} as follows: 1) use n_h, w_h, and c to reconstruct f_Θ; 2) dequantize $\breve{\Theta}$ to $\tilde{\Theta}$ and use it to initialize the parameters of f_Θ; 3) use the width w and height h to set up the input grid between $[-1,+1] \times [-1,+1]$; and 4) evaluate $f_{\tilde{\Theta}}$ at each location in the input grid to reconstruct the image $\tilde{I}$ (see Fig. 1).

IV. Experimental Setup and Ablative Study

This section discusses the datasets that we have used to study and evaluate the proposed method. It also discusses the various design aspects—architecture search, compression and decompression times, model fitting, and the effects of random sampling—of the proposed model. We evaluate the proposed method against other schemes in Section V.

We have used four datasets to evaluate our approach. We have selected these datasets since others have used these previously to study hyperspectral image compression. Furthermore, these datasets have been extensively used in the classification and analysis of hyperspectral imagery. Indian Pines, Cuprite, and Jasper Ridge datasets have been collected using NASA's Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The AVIRIS sensor gathers geometrically coherent spectroradiometric data that can be used to characterize the Earth's surface. The Pavia University dataset was captured using the Reflective Optics System Imaging Spectrometer (ROSIS).

Indian Pines is a $145 \times 145 \times 220$ hyperspectral image. It was collected using the AVIRIS sensor in 1992, and it spans a region over NW Indiana. The hyperspectral image contains a mix of farms and wooded areas. In addition, the image contains low-density built-up regions, houses, a number of secondary roads, a rail line, and two dual-lane motorways. **Jasper Ridge** is a $100 \times 100 \times 224$ hyperspectral image. It was also captured using the AVIRIS sensor. **Pavia University** is a $610 \times 340 \times 103$ hyperspectral image. This dataset was captured by the ROSIS-03 aerial instrument in 2001. The dataset was acquired from the University of Pavia. The **Cuprite** dataset contains one $614 \times 512 \times 224$ hyperspectral image.

A. Architecture Search

Given an image and our (MLP) parameter budget, which is measured in bits per pixel per band, or bpppb, the first goal is to select the MLP structure—i.e., the number of hidden layers and their widths—that is able to represent this image with an acceptable PSNR value. Fig. 4 shows PSNR values achieved for different architectures for the four datasets having a fixed bpppb budget. This suggests that network structure, in addition to network capacity, affects how well a network represents the hyperspectral image.

The MLP structure is chosen via hyperparameter search, which involves training feasible designs containing the right number of hidden layers having the correct width on a given hyperspectral image. The result of this process is a single MLP that is able to reconstruct the hyperspectral image with the

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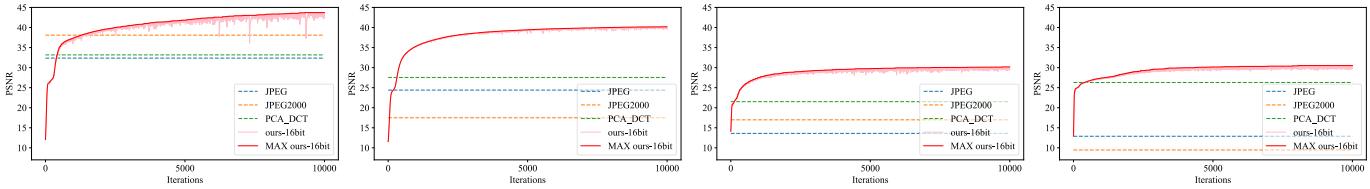


Fig. 5. Encoding procedure. Model training on (from left to right) Indian Pines at 0.2 bpppb, Jasper Ridge at 0.15 bpppb, Pavia University at 0.025 bpppb, and Cuprite at 0.02 bpppb. At around 2000 iteration mark, our method is already achieving better PSNR values than those for JPEG, JPEG2000, and PCA-DCT. Furthermore, the PSNR value for our methods continues to improve with more iterations (up to a point). (a) Indian Pines. (b) Jasper Ridge. (c) Pavia University. (d) Cuprite.

TABLE II

EFFECT OF SAMPLING ON COMPRESSION TIMES. PREFIX "OURS" DENOTES THE METHODS PROPOSED HERE. OURS ARE LEARNING-BASED METHODS, AND AS EXPECTED, THESE POST SOME OF THE HIGHEST COMPRESSION TIMES. WE ADDRESS THIS ISSUE SOMEWHAT THROUGH "SAMPLING," WHICH HALVES THE COMPRESSION TIMES. FOR THESE RESULTS, THE SAMPLING RATE IS CHOSEN TO BE 20%. THE GOOD NEWS IS THAT "OURS" METHODS ACHIEVE DECOMPRESSION TIMES THAT ARE FASTER THAN THOSE FOR JPEG, JPEG2000, AND PCA-DCT SCHEMES. THESE RESULTS CONFIRM THAT: 1) SAMPLING HELPS AND 2) THE PROPOSED METHOD IS WELL-SUITED TO "COMPRESS-ONCE" SORT OF APPLICATIONS

Dataset	Method	bppppb	compression time (Sec)	decompression time (Sec)	PSNR ↑
Indian Pines	JPEG$^+$ [21], [22]	0.1	7.353	3.27	27.47
	JPEG2000$^+$ [23]	0.1	0.1455	0.3115	33.58
	PCA-DCT$^+$ [24]	0.1	1.66	0.04	32.28
	ours-32bit	0.1	243.64	0	36.98
	ours-16bit	0.05	243.64	0	36.95
	ours-sampling-32bit	0.1	282.08	0.0005	**40.1**
	ours-sampling-16bit	0.05	282.08	0.0005	28.40
Jasper Ridge	JPEG$^+$ [21], [22]	0.1	3.71	1.62	21.13
	JPEG2000$^+$ [23]	0.1	0.138	0.395	17.49
	PCA-DCT$^+$ [24]	0.1	1.029	0.027	26.82
	ours-32bit	0.1	312.38	0.0005	32.54
	ours-16bit	0.06	312.38	0.0005	32.51
	ours-sampling-32bit	0.1	75.91	0.0005	**34.77**
	ours-sampling-16bit	0.06	75.91	0.0005	22.07
Pavia University	JPEG$^+$ [21], [22]	0.1	33.86	14.61	20.25
	JPEG2000$^+$ [23]	0.1	0.408	0.628	17.75
	PCA-DCT$^+$ [24]	0.1	6.525	0.235	25.43
	ours-32bit	0.1	780.16	0.0009	34.46
	ours-16bit	0.05	780.16	0.0009	34.17
	ours-sampling-32bit	0.1	72.512	0.0004	**38.08**
	ours-sampling-16bit	0.05	72.512	0.0004	27.02
Cuprite	JPEG$^+$ [21], [22]	0.06	101.195	45.02	12.88
	JPEG2000$^+$ [23]	0.06	1.193	2.476	15.16
	PCA-DCT$^+$ [24]	0.06	11.67	0.754	26.75
	ours-32bit	0.06	1565.97	0.001	28.02
	ours-16bit	0.03	1565.97	0.001	27.90
	ours-sampling-32bit	0.06	664.87	0.001	**37.27**
	ours-sampling-16bit	0.03	664.87	0.001	24.85

desired PSNR value. The parameters of the final MLP are then quantized to 16-bit (or 8-bit) precision, which leads to further savings in terms of the storage needed to represent the hyperspectral image. Our experiments suggest that reducing the MLP parameters from 32- to 16-bit precision did not increase distortion, and it had little effect on the signal-to-noise ratio. However, reducing these parameters to 8-bit precision had an adverse effect.

B. Encoding Considerations

Our method belongs to the class of "slow-encoding-fast-decoding" compression methods. The method needs to train, actually *overfit*, multiple MLPs at encoding (compression) time. This is needed to find the MLP structure that best represents the hyperspectral image given a particular storage budget. Decoding, however, only requires evaluating this MLP at various pixel locations. Decoding is fast. It can be made even faster by exploiting the parallelism inherent to this procedure. The "slow-encoding-fast-decoding" nature of this method makes it well-suited for applications where the hyperspectral image is compressed once only, say at capture time. It is obvious that encoding is also compute-heavy, and that is something we need to keep in mind as we imagine hyperspectral sensors capable of compressing hyperspectral images at capture time using the method proposed here.

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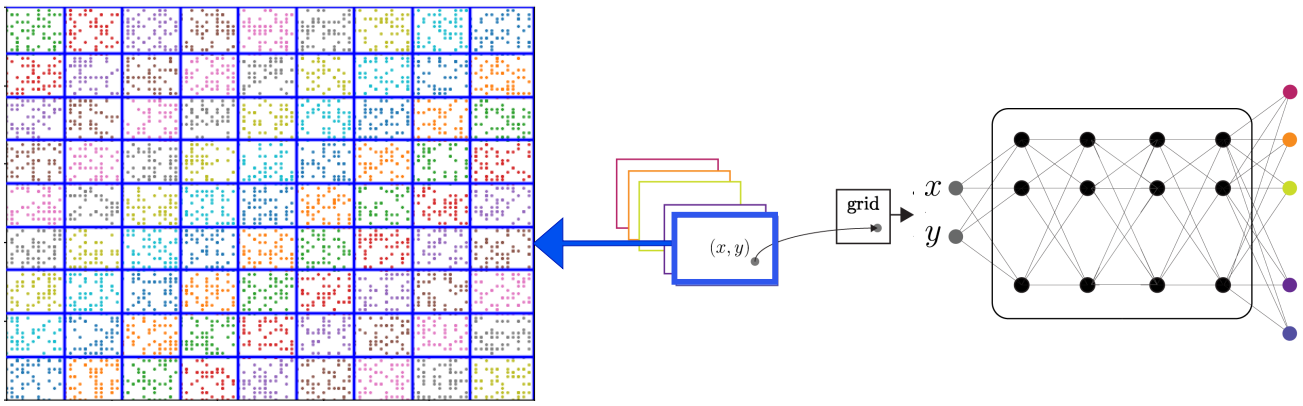


Fig. 6. Compression with random sampling. Pixel locations are uniformly sampled during model training. This is a departure from the traditional approach of using all pixel locations when fitting INRs.

TABLE III

RESULTS FOR FIXED bpppb. COMPRESSION RATES (I.E., THE DESIRED SIZE OF THE COMPRESSED DATA) ARE FIXED ACROSS METHODS. THE QUALITY OF COMPRESSION IS EXPRESSED IN TERMS OF PSNR. FIVE VARIANTS OF THE PROPOSED METHODS ARE INCLUDED; THESE ARE INDICATED BY THE PREFIX "OURS." CLASSICAL AND LEARNING-BASED METHODS ARE DENOTED USING $^+$ AND † TEXT DECORATIONS. THE LAST COLUMN INDICATES THE NETWORK STRUCTURE: n_h AND w_h INDICATE THE NUMBER OF HIDDEN LAYERS AND THE WIDTH OF THE HIDDEN LAYERS, RESPECTIVELY. THIS COLUMN APPLIES TO OUR METHODS ONLY. THE BEST PSNR VALUES FOR EACH DATASET ARE SHOWN IN BOLD. QUANTIZATION TO 16 OR 8 bits REDUCES THE bpppb BY ONE-HALF AND ONE-FOURTH, RESPECTIVELY. RESULTS FOR THE VIDEO- AND AUTOENCODER-BASED SCHEMES ARE MISSING, SINCE THESE METHODS DO NOT PROVIDE RESULTS FOR THE bpppb VALUES USED HERE, RESULTS FOR VIDEO-BASED METHOD ARE AVAILABLE IN TABLE IV, AND THE RESULTS FOR AUTOENCODER-BASED METHOD ARE PLOTTED IN FIG. 7

Method	Dataset	Size (KB)	PSNR	bpppb	n_h, w_h	Dataset	Size (KB)	PSNR	bpppb	n_h, w_h
-	Indian Pines	9251	∞	16	-,-	Jasper Ridge	4800	∞	16	-,-
JPEG$^+$ [21], [22]		115.6	34.085	0.2	-,-		30	21.130	0.1	-,-
JPEG2000$^+$ [23]		115.6	35.84	0.2	-,-		30	17.494	0.1	-,-
PCA-DCT$^+$ [24]		115.6	33.173	0.2	-,-		30	26.821	0.1	-,-
3D-SPECK$^+$ [35]		115.6	-	0.2	-,-		30	-	0.1	-,-
PCA-JPEG2000† [25]–[27]		115.6	39.5	0.2	-,-		30	**44.26**	0.1	-,-
FPCA-JPEG2000† [29]		115.6	40.5	0.2	-.-		30	-	0.1	-,-
3D-DCT† [30]		115.6	-	0.2	-,-		30	-	0.1	-,-
3D-DWT-SVR† [31]		115.6	-	0.2	-,-		30	-	0.1	-,-
WSRC† [32]		115.6	-	0.2	-,-		30	-	0.1	-,-
ours-32bit		115.6	39.10	0.2	5,60		30	32.54	0.1	5,20
ours-16bit		57.5	38.99	0.1	5,60		15	32.51	0.06	5,20
ours-sampling-32bit		115.6	**42.22**	0.2	5,60		30	34.77	0.1	5,20
ours-sampling-16bit		57.5	29.68	0.1	5,60		15	22.07	0.06	5,20
ours-sampling-8bit		28.7	32.97	0.05	5,60		7.5	24.32	0.03	5,20
-	Pavia University	42724	∞	16	-,-	Cuprite	140836	∞	16	-,-
JPEG$^+$ [21], [22]		267	20.253	0.1	-,-		880.2	24.274	0.1	-,-
JPEG2000$^+$ [23]		267	17.752	0.1	-,-		880.2	20.889	0.1	-,-
PCA-DCT$^+$ [24]		267	25.436	0.1	-,-		880.2	27.302	0.1	-,-
3D-SPECK$^+$ [35]		267	-	0.1	-,-		880.2	27.1	0.1	-,-
PCA-JPEG2000† [25]–[27]		267	33	0.1	-,-		880.2	**40.90**	0.1	-,-
FPCA-JPEG2000† [29]		267	-	0.1	-,-		880.2	-	0.1	-,-
3D-DCT† [30]		267	-	0.1	-,-		880.2	33.4	0.1	-,-
3D-DWT-SVR† [31]		267	-	0.1	-,-		880.2	28.20	0.1	-,-
WSRC† [32]		267	-	0.1	-,-		880.2	35	0.1	-,-
ours-32bit		267	34.46	0.1	10,80		880.2	28.954	0.1	25,100
ours-16bit		133.5	34.17	0.05	10,80		440.1	24.334	0.06	25,100
ours-sampling-32bit		267	**38.08**	0.1	10,80		880.2	36.55	0.1	25,90
ours-sampling-16bit		133.5	27.02	0.05	10,80		440.1	24.91	0.06	25,90
ours-sampling-8bit		66.75	24.02	0.02	10,80		220.05	22.35	0.03	25,90

We show an example of the overfitting procedure in Fig. 5. These plots show the encoding procedure on the four datasets: 1) Indian Pines at 0.2 bpppb; 2) Jasper Ridge at 0.15 bpppb; 3) Pavia University at 0.025 bpppb; and 4) Cuprite at 0.02 bpppb. JPEG, JPEG2000, and PCA-DCT methods do not require iterations. Consequently, their respective PSNR values

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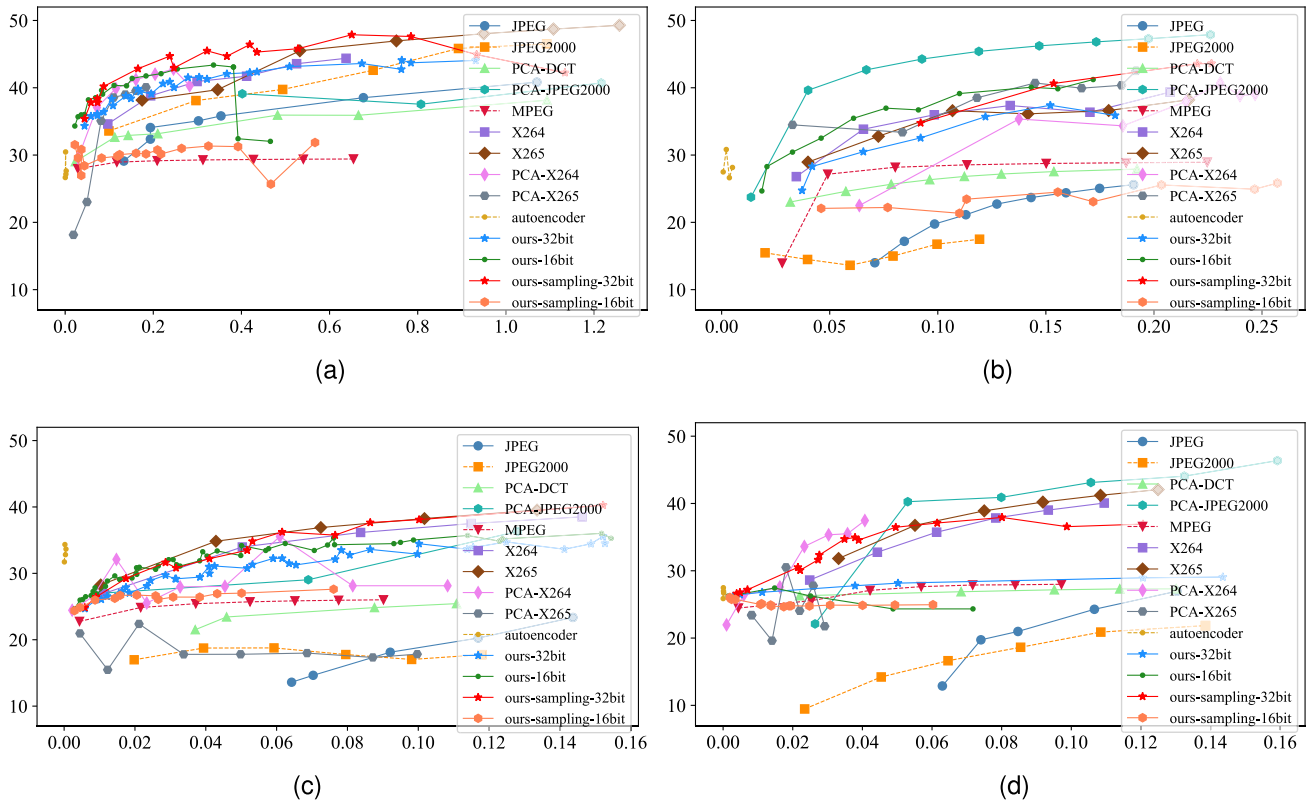


Fig. 7. PSNR values versus bpppb. The x-axis represents bpppb values, and the y-axis represents PSNR values. The uncompressed hyperspectral images have a bpppb value of 16. This figure is best viewed in color. (a) Indian Pines. (b) Jasper Ridge. (c) Pavia University. (d) Cuprite.

are denoted with the horizontal dashed lines. The method proposed in this article is iterative. Note that PSNR values for *ours-16 bit* continue to increase with the number of iterations (up to a point). Improvement in PSNR values saturates at around 10k, 15k, 10k, and 5k iterations for Indian Pines, Jasper Ridge, Pavia University, and Cuprite datasets, respectively. This hints at the upper bound on encoding, or compression, time for our method. Note also that at around 2k iteration mark, *ours-16 bit* method starts to obtain better PSNR values than the other three methods. As stated earlier, our method involves model fitting, which is inherently stochastic. Therefore, throughout the iterative process, we store the model parameters that obtained the highest value for PSNR thus far. In these plots, *MAX ours-16 bit* denotes these PSNR scores. This guarantees that the model does not get worse over time.

C. Model Fitting

The number of inputs for all our models is 2, and the number of outputs is equal to the number of channels (or bands) of the hyperspectral image. The activation function for hidden layers is sinusoidal. We initialized the MLP using the guidelines provided in [55]. The Adam optimizer was used during training, and the learning rate was set to $2e-4$. All experiments were conducted on an Intel i7 desktop with Nvidia RTX 2080 GPU.

D. Random Sampling

During compression, the INR model is trained by iterating over $w \times h$ pixel locations for each epoch. For hyperspectral images with high-spatial resolution, this leads to longer compression times. Hyperspectral images, similar to color images, exhibit spatial coherence. We use this assumption and explore the effect of pixel location sampling, hereafter referred to as sampling, during INR model training. The idea is to sample a fraction of pixel locations at training time. Fig. 6 illustrates the sampling procedure. First, the image is divided into nonoverlapping tiles (in the spatial domain). This is done in order to ensure that the same fraction of pixel locations is sampled from each region of the image. The colored dots indicate the sampled locations within each tile. The sampling is performed with rejection to prevent the case where the same pixel location is sampled multiple times during each epoch. Furthermore, there is no restriction that the same pixel locations are sampled across epochs. The sampling procedure is controlled by two parameters: 1) tiles and 2) the fraction of locations to be sampled within each tile. For the results presented in Table II, the image is divided into 9×9 tiles, and the sampling fraction is set to 20%. This table shows compression and decompression times for four datasets. As expected, sampling reduces the compression times in half for all datasets. What is more intriguing, however, is that for comparable bpppb values, sampling results in higher PSNR values. We do not yet fully understand why this is so; *ours-sampling-32 bit* and *ours-sampling-16 bit* denote models that used sampling during training. As earlier, prefix *HP* denotes that model parameters are stored using 16-bit precision.

V. Results

We compare our method with a number of: 1) classical; 2) learning-based; and 3) video-based schemes. Table III lists

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compression results in terms of PSNR with a number of classical and learning-based hyperspectral image compression approaches. In order to compare the performance of different methods, the target bpppb is fixed across methods. For the Indian Pines dataset, the target bpppb is set to 0.2, and for the other three datasets, it is set to 0.1. These values are selected since results were available for other methods for these values. Note that the uncompressed bpppb for all datasets is 16. Results for other methods are taken from their respective publications—Cuprite is a popular dataset since most of the other methods have results available for this dataset. The last column describes the network structure used for INR learning: n_h and w_h refer to the number of hidden layers and the width of these layers, respectively. The proposed method achieves better PSNR than other methods for Indian Pines and Pavia University datasets. However, the PCA-JPEG2000 method posts better PSNR than our methods for Jasper Ridge and Cuprite datasets.

A. Compression Rates

Fig. 7 shows PSNR values at various compression rates for different methods. Specifically, we compare our method, denoted by the prefix "ours" against JPEG, JPEG2000, PCA-DCT, PCA-JPEG2000, MPEG, X264, X265, PCA-X264, PCA-X265, and an autoencoder-based method [36]. This figure does not include the result for *ours-*-8 bit* since our primary focus has been on 32- and 16-bit precisions. This is in part due to the fact that storing network parameters in 8-bit precision adversely affects the compression results. The plots suggest that the proposed method achieves higher compression quality at lower bpppb values (i.e., higher compression rates).

For the Indian Pines dataset, *our-sampling-32 bit* achieves better PSNR up to around 0.7 bpppb, at which point X265 obtains better PSNR. What is curious is that the PSNR for *ours-16 bit* drops drastically at around 0.4 bpppb. This merits further investigation. The *ours-sampling-32 bit* method got better PSNR values than all other methods at various bpppb levels. For Jasper Ridge, *ours-sampling-32 bit* performs better than other methods except for PCA-JPEG2000, which achieves better PSNR for all bpppb values. For Pavia University, the *ours-sampling-32 bit* method achieves PSNR that is better than every method except for X265, whose results are comparable to the proposed method. For Cuprite, *ours-sampling-32 bit* is in the top four methods. It achieves the PSNR at lower bpppb values. These results confirm that the proposed model is among the top methods across the four datasets at various bpppb's. In addition, the proposed method seems well-suited for lower values of bpppb.

B. Video-Based Methods

Table IV evaluates *ours-sampling-8 bit* against seven video-based methods that treat various channels of a hyperspectral image as frames of a video and employ video coding techniques to achieve compression.[2] The bpppb is fixed at 0.1 for Indian Pines, 0.15 for Jasper Ridge, 0.1 for Pavia University, and 0.03 for the Cuprite dataset. The best results are shown in bold. Our method outperforms all other methods on Indian Pines, Jasper Ridge, and Pavia University datasets, and our method achieves the second-best PSNR value on the Cuprite dataset. These results also confirm that "sampling," which not only reduces encoding times, achieves top PSNR scores.

TABLE IV
COMPARING THE PROPOSED METHOD (WITH SAMPLING RATE OF 20%) AGAINST VIDEO-BASED SCHEMES. OUR METHOD ACHIEVES BETTER RESULTS THAN ALL OTHER METHODS ON INDIAN PINES, JASPER RIDGE, AND PAVIA UNIVERSITY DATASETS. IN ADDITION, OUR METHOD ACHIEVES THE SECOND BEST PSNR VALUE AFTER PCA-X264 ON THE CUPRITE DATASET. TEXT DECORATION ‡ INDICATES A VIDEO-BASED METHODS

Dataset	Method	bpppb	PSNR ↑
Indian Pines	X264‡ [25]–[27]	0.1	34.61
	X265‡ [25]–[27]	0.1	38.5
	PCA-X264‡ [25]–[27]	0.1	39.8
	PCA-X265‡ [25]–[27]	0.1	38.1
	MPEG‡ [28]	0.1	28.9
	HEVC‡ [33]	0.1	30
	RPM‡ [34]	0.1	31
	ours-sampling-32bit	0.1	**40.1**
Jasper Ridge	X264‡ [25]–[27]	0.15	37.35
	X265‡ [25]–[27]	0.15	36.12
	PCA-X264‡ [25]–[27]	0.15	35.35
	PCA-X265‡ [25]–[27]	0.15	39.94
	MPEG‡ [28]	0.15	28.75
	HEVC‡ [33]	0.15	-
	RPM‡ [34]	0.15	-
	ours-sampling-32bit	0.15	**41.15**
Pavia University	X264‡ [25]–[27]	0.1	37.17
	X265‡ [25]–[27]	0.1	37.90
	PCA-X264‡ [25]–[27]	0.1	28.13
	PCA-X265‡ [25]–[27]	0.1	17.82
	MPEG‡ [28]	0.1	26.01
	HEVC‡ [33]	0.1	-
	RPM‡ [34]	0.1	-
	ours-sampling-32bit	0.1	**38.08**
Cuprite	X264‡ [25]–[27]	0.03	28.6
	X265‡ [25]–[27]	0.03	31.8
	PCA-X264‡ [25]–[27]	0.03	**35.5**
	PCA-X265‡ [25]–[27]	0.03	21.7
	MPEG‡ [28]	0.03	25.5
	HEVC‡ [33]	0.03	25
	RPM‡ [34]	0.03	29
	ours-sampling-32bit	0.03	34.9

C. SSIM Comparison

Table V uses SSIM metric to compare our method on Cuprite and Pavia University datasets against other schemes for fixed bpppb. The results reported for other methods are taken from their respective publications. SSIM scores for other methods for Indian Pines and Jasper Ridge datasets are not available; therefore, this table does not include Indian Pines and Jasper Ridge datasets. Nevertheless, these results suggest that the proposed scheme is able to preserve "perceptual quality" in the compressed signal.

For the Cuprite dataset, we compare our method against WSRC [32], 3D-SPECK [35], 3D-SPHIT [67], 3D-WBTC [68], 3D-LSK [35], 3D-NLS [69], 3D-LMBTC [70],

[2]It is not possible to include the results presented in Table IV since the bpppb values used by video-based methods do not match those used by methods listed in Table III.

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TABLE V
SSIM COMPARISON FOR THE CUPRITE AND PAVIA UNIVERSITY DATASETS AT FIXED bpppb VALUES. VALUES FOR OTHER METHODS ARE TAKEN FROM THEIR RESPECTIVE PUBLICATIONS. CONSEQUENTLY, WSRC IS EVALUATED AT bpppb = 0.1 FOR THE CUPRITE DATASET, AND ALL OTHER METHODS ARE EVALUATED FOR bpppb = 0.1. AT THE TIME OF WRITING, THE SSIM SCORES WERE NOT AVAILABLE FOR OTHER METHODS ON INDIAN PINES AND JASPER RIDGE DATASETS

Dataset	bpppb	Method	SSIM ↑
Cuprite	0.1	WSRC [32]	0.75
		ours-sampling-32bit	**0.9798**
	0.01	3D-SPECK [35]	0.142
		3D-SPIHT [68]	0.136
		3D-WBTC [69]	0.141
		3D-LSK [35]	0.138
		3D-NLS [70]	0.135
		3D-LMBTC [71]	0.140
		3D-ZM-SPECK [72]	0.141
		ours-32bit	**0.9565**
		ours-16bit	0.9514
		ours-sampling-32bit	0.9527
		ours-sampling-16bit	0.9390
Pavia University	0.1	3D-SPHIT [68]	0.4
		3D-DCT [30]	0.85
		ours-32bit	0.9564
		ours-16bit	0.9527
		ours-sampling-32bit	**0.9901**
		ours-sampling-16bit	0.8518

and 3D-ZM-SPECK [71]. WSRC is evaluated at bpppb = 0.1, and the *ours-sampling-32 bit* method outperforms WSRC at this bpppb on the Cuprite dataset. 3D-SPECK, 3D-SPHIT, 3D-WBTC, 3D-LSK, 3D-NLS, 3D-LMBTC, and 3D-ZM-SPECK are evaluated at bpppb = 0.01, and the *ours-32 bit* method achieves better SSIM scores than these methods at this bpppb.

On the Pavia University dataset, we compare our method against 3D-SPHIT and 3D-DCT [30] methods. These methods are evaluated at bpppb = 0.1, and the *ours-sampling-32 bit* method achieves the highest SSIM score for this bpppb value.

VI. CONCLUSION

We employ INRs to compress hyperspectral images. Multilayer perceptron neural networks with sinusoidal activation layers are overfitted to a hyperspectral image. The network is trained to map pixel locations to pixels' spectral signatures. The parameters of the network, along with its structure, represent a compressed encoding of the original hyperspectral image. We also use a sampling method that is controlled by: 1) window size and 2) sampling rate to reduce the compression time. We have evaluated our approach on four benchmarks, and we draw the following conclusions: 1) the proposed method (with/without sampling; 32-/16-bit quantization) performs high-quality compression at high compression rates; 2) it is beneficial to perform architecture search and examine the effects of quantization at compression time since, in some datasets, *ours-*-16 bit* outperforms *ours-*-32 bit*; 3) the compression quality obtained by the proposed method is better or in some cases similar to that achieved by other schemes; and 4) sampling improves compression times while maintaining acceptable reconstruction quality as measured by PSNR. Being a learning-based method that requires overfitting an MLP to data at compression time, the proposed method falls into the category of "slow-encoding-fast-decoding" schemes. In the future, we plan to investigate techniques to speed up the encoding process and reduce the computational requirements.

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